

Data Scientist and Python Developer with 4+ years of experience designing and implementing data-driven solutions. Strong foundation in Machine Learning, Data Science, and Artificial Intelligence (Master's, Aalto University). Skilled in Big Data analysis, Natural Language Processing (NLP), Time Series modeling, Data Visualization, algorithm optimization, and applied Generative AI using Large Language Models (LLMs). Passionate about developing scalable, practical solutions that leverage modern GenAI techniques to drive measurable business impact. Firm believer that simplicity and efficiency often outweigh unnecessary complexity.

EDUCATION

Masters in Machine Learning, Artificial Intelligence and Data Science , <i>Aalto University</i> (current GPA: 4.34/5.0)	2021 — 2025
B.Tech in Electronics and Computer Engineering , <i>Vellore Institute of Technology</i> (CGPA: 9.01/10.00)	2016 — 2020

SKILLS

Programming & Tools	Python, SQL, C/C++, AWS, Azure, MySQL, Power BI
ML & Data Science	Machine Learning, NLP, Time Series Modeling, Geospatial Analytics, Data Visualization
Deep Learning & Generative AI	PyTorch, Transformers, LangChain, LangGraph, FAISS, RAG, LLMs, Ollama

TECHNICAL EXPERIENCE

DATA SCIENTIST <i>SkenarioLabs Oy</i>	Sept 2023 — Present <i>Espoo, Finland</i>
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- Developed advanced regression and predictive models for real estate valuation, feature forecasting, and market trend analysis.
- Conducted climate risk (physical risk) analysis to assess the impact of environmental factors on property and portfolio performance.
- Worked extensively with GeoPandas and geospatial data (vector polygons, coordinate reference systems) for spatial modeling and visualization.
- Designed and deployed interactive dashboards using Python's Plotly and Dash libraries for data exploration and reporting.
- Collaborated using modern development tools including Bitbucket, JIRA, and AWS for version control, project management, and cloud integration.

RESEARCH ASSISTANT <i>Aalto University</i>	Apr 2023 — Sept 2023 <i>Espoo, Finland</i>
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- Conducted research for my thesis titled *Spatio-Temporal Video Grounding*.
- Explored various video datasets and developed methods for their analysis.
- Designed and implemented deep learning models using Transformer architectures in PyTorch.

DATA SCIENTIST <i>Pricen</i>	May 2022 — Mar 2023 <i>Helsinki, Finland</i>
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- Developed data visualizations in Python for generating automated reports.
- Built machine learning models to predict and suggest prices for revenue optimization.

SOFTWARE ENGINEER <i>Stealth Mode Quantitative Macro Asset Investment Company</i>	May 2021 — Oct 2021 <i>San Francisco, USA</i>
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- Conducted data science and statistical analysis.
- Developed models using Python, NumPy, and SciPy; studied global market dynamics.

ASSOCIATE SOFTWARE ENGINEER AND PYTHON DEVELOPER INTERN <i>Bajaj Finserv Health Limited</i>	Jan 2020 — Apr 2021 <i>Pune, India</i>
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- Performed customer claims and patient journey analyses to identify key business insights.
- Gained hands-on experience with AWS and Azure cloud platforms.
- Implemented speech recognition solutions to facilitate voice prescriptions and reduce turnaround time.
- Developed and maintained dashboards in Power BI for data visualization and reporting.
- Conducted What-If and sensitivity analyses for claims evaluation.
- Worked on API development, integration, and production deployment.
- Contributed to multiple projects utilizing Python and SQL.

SUMMER INTERN <i>Nokia Solutions and Networks Pvt. Ltd.</i>	May 2019 — Jun 2019 <i>Chennai, India</i>
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- Implemented a "Pick-To-Light" tool to assist workers in error-free equipment packing.
- Utilized the OpenCV Python library and a machine learning model for system development.
- Learned the fundamentals of Vuforia, a widely used platform for augmented reality development.

SUMMER RESEARCH INTERN <i>IIT Madras</i>	Jun 2018 — Jul 2018 <i>Chennai, India</i>
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- Gained exposure to fundamental concepts of deep learning and neural networks.
- Built and evaluated various classification models using MATLAB.
- Utilized CIFAR and MNIST datasets for image classification experiments.

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Hans Tiwari

Data Scientist

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Github

PUBLICATIONS

- **Size-Modulated Deformable Attention in Spatio-Temporal Video Grounding Pipelines** (ICPR '24)
- **Spatio-Temporal Video Grounding of Human Actions for Human-in-the-Loop Artificial Intelligence** (CAIHU '24)
- **Design of Kids-Specific URL Classifier using Recurrent Convolutional Neural Network** (ICCIDS '19, Procedia Computer Science)
- **Bidirectional GRU-Based Attention Model for Kid-Specific URL Classification** (IGI Global '19)
- **Internet of Vehicles (IoV) Based Voice Controlled Autonomous Vehicle for Surveillance** (IJRTE '19)

RELEVANT PROJECTS

- MASTER'S THESIS: SPATIO-TEMPORAL VIDEO GROUNDING USING TRANSFORMERS** Sept 2023
- Conducted a Master's thesis on Spatio-Temporal Video Grounding (STVG), focusing on aligning video segments with natural language descriptions.
 - Modified Transformer-based STVG models to better handle language ambiguity.
 - Analyzed attention mechanisms, anchor queries, and object scale sensitivity.
- DOMAIN-SPECIFIC KNOWLEDGE ASSISTANT (RAG)** 2024
- Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) system for domain-specific question answering.
 - Built document ingestion, chunking, and vector-based retrieval pipelines.
 - Deployed the system as a production-style API with grounded LLM responses.
- LLM-POWERED ANALYTICS ASSISTANT** 2024
- Developed an LLM-driven analytics assistant for querying structured tabular data using natural language.
 - Implemented a planning-execution framework combining LLM reasoning with deterministic Python analysis.
 - Generated human-readable explanations of analytical results.
- BERT-BASED QUESTION ANSWERING USING SQuAD** 2023
- Implemented and fine-tuned BERT-based models for extractive question answering.
 - Trained models on the SQuAD dataset for span prediction tasks.
 - Evaluated performance using Exact Match (EM) and F1 metrics.
- VIVIT FOR VIDEO ACTION RECOGNITION** Autumn 2022
- Evaluated Transformer-based ViViT architectures for video action recognition.
 - Trained and tested models on ActivityNet and UCF101 datasets.
 - Analyzed factorized encoders for effective spatio-temporal feature learning.